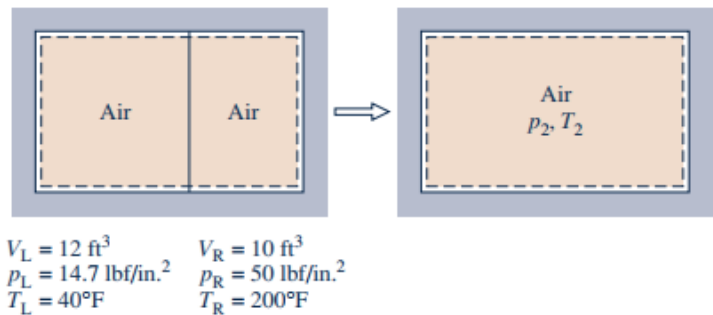
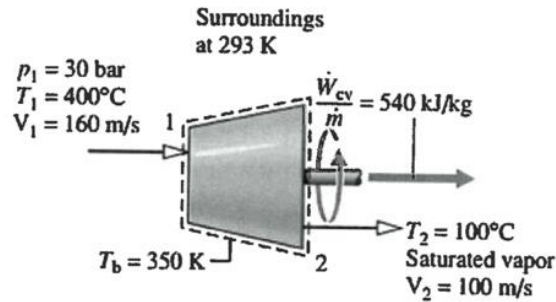


Assignment # 8

A rigid, well-insulated tank contains air. A partition in the tank separates 12 ft^3 of air at 14.7 lbf/in.^2 , 40°F from 10 ft^3 of air at 50 lbf/in.^2 , 200°F , as illustrated in Fig. below. The partition is removed and air from the two sides mix until a final equilibrium state is attained. The final temperature reaches 149.3°F . The air can be modeled as an ideal gas, and kinetic and potential energy effects can be neglected. Determine (a): The final pressure, in lbf/in.^2 (b): The amount of entropy produced, in $\text{Btu/}^\circ\text{R}$.



7. (25)-Steam enters a turbine with a pressure of 30 bar, a temperature of 400C, and a velocity of 160 m/s. Saturated vapor at 100C exits with a velocity of 100 m/s. At steady state, the turbine develops work equal to 540 kJ per kg of steam flowing through the turbine. Heat transfer between the turbine and its surroundings occurs at an average outer surface temperature of 350 K. Determine the rate at which entropy is produced within the turbine per kg of steam flowing, in kJ/kg K. Neglect the change in potential energy between inlet and exit. Show the following steps in your calculations:



- (5)-Summarizes the steady-state form of the entropy rate balance equation for the given Control Volumes as shown on the figure.
- (5)-Determine the value of H_1 (inlet) and H_2 (exit) at given pressure and temperature.
- (5)-Determine the value of S_1 (inlet) and S_2 (exit) at given pressure and temperature.
- (5)-Determine the heat transfer rate, $\frac{\dot{Q}_{cv}}{\dot{m}}$ required for the process.
- (5)-Determine the entropy production per unit mass $\frac{\dot{\sigma}_{cv}}{\dot{m}}$ flowing through the turbine.